

Why rare detections are hard to learn

Only 5 detections in 10,000 events

At a 0.05% detection rate, a network can look excellent while missing every positive event.

Three different problems

Misleading accuracy

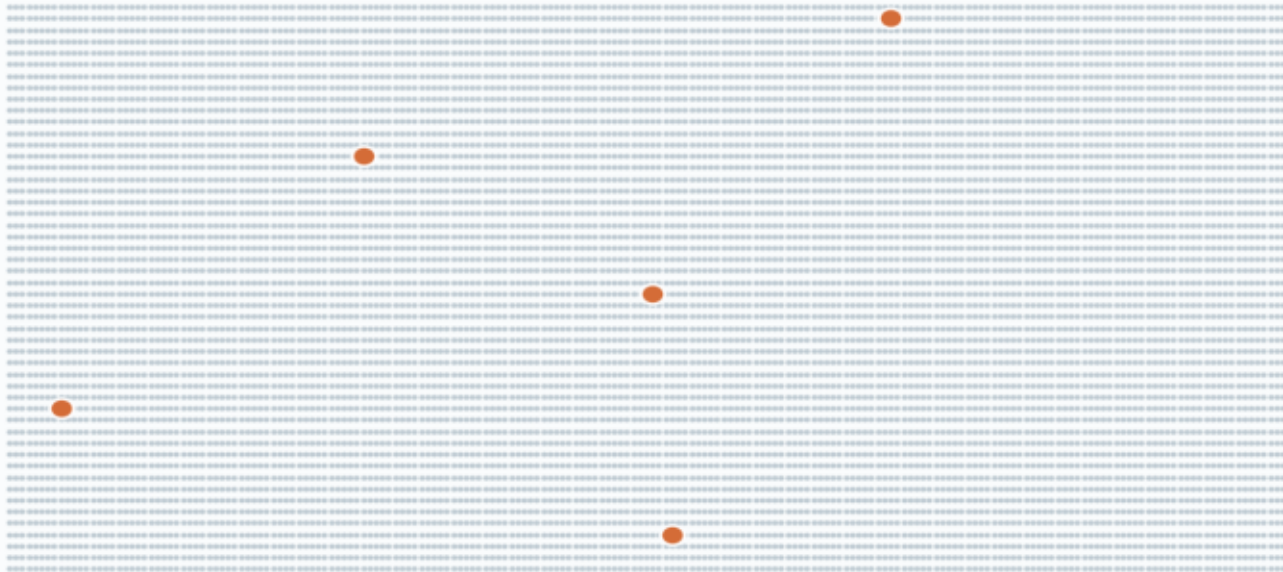
Sparse evidence

Shifted probabilities

The goal: learn rare-event structure without distorting the population rate.

99.95% accuracy — and zero detections found

10,000 events: 9,995 negatives + 5 positives



Predict “negative” for all

Accuracy: 99.95%

Positive recall: 0%

Every detection is missed.

Accuracy counts successes mostly on the abundant negative class.

A negative decision is not a zero probability

Classification decision

“Does this event produce a hit?”

If the threshold is 50%, a score of 0.05% is classified as negative.

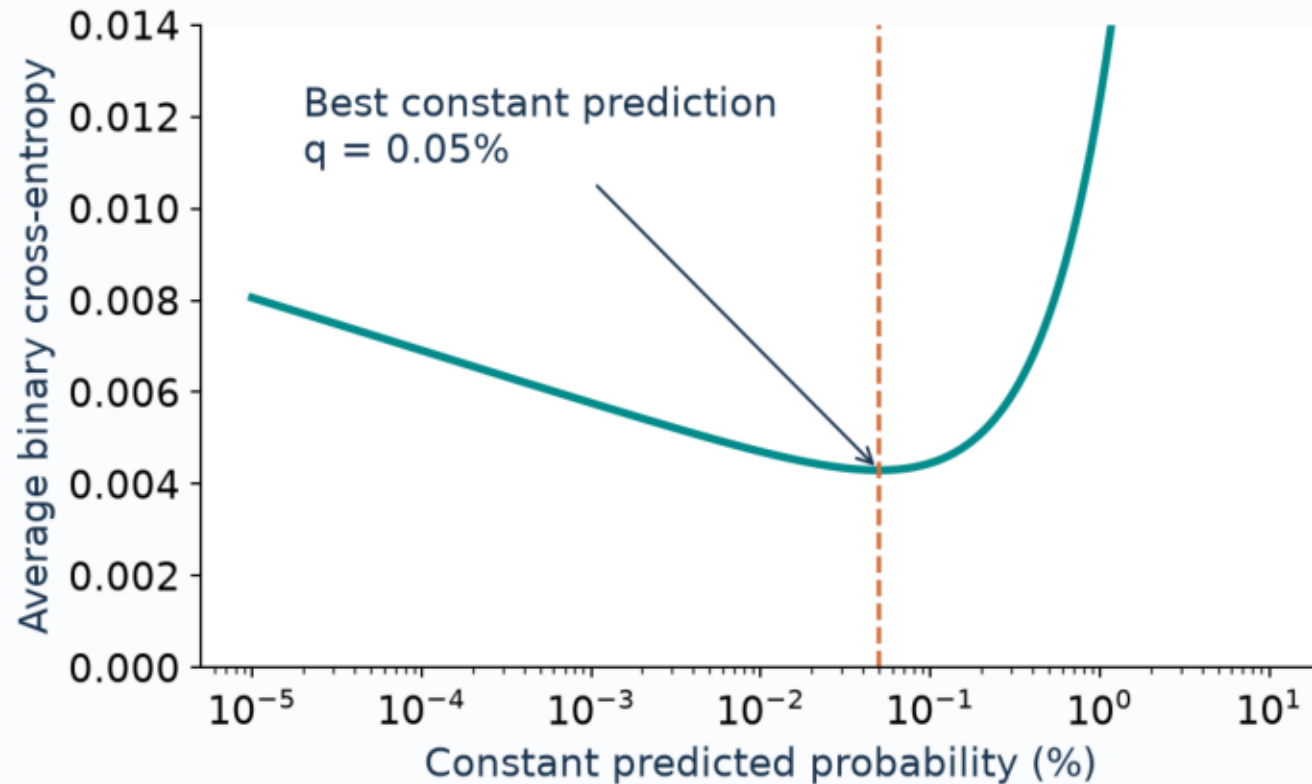
Probability estimate

“How likely is a hit?”

A score of 0.05% for all events predicts 5 hits per 10,000 on average.

A model can have zero recall at threshold 0.5 and still estimate the mean rate.

BCE does not reward predicting exactly zero



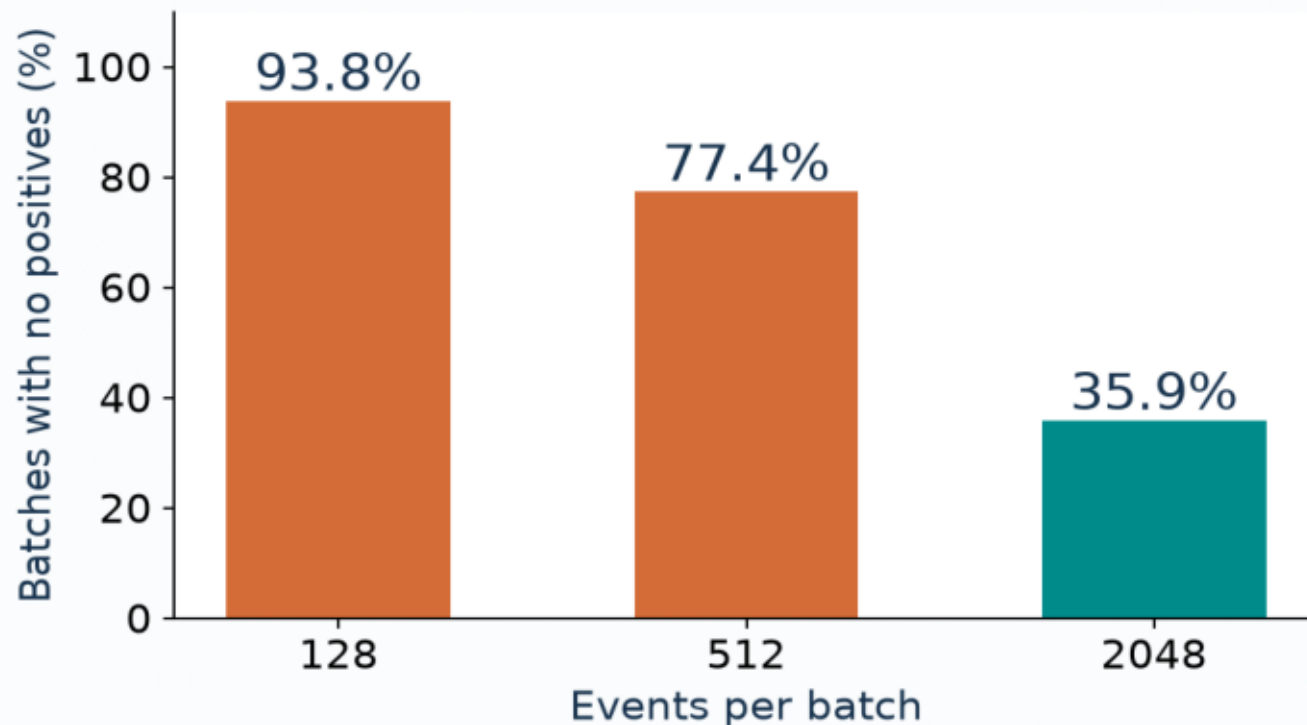
A positive event has loss $-\log(q)$.

As q approaches zero, that penalty grows without bound.

The best constant score is the true positive fraction.

Near-zero scores can be sensible; failing to learn useful variation is the concern.

Many small batches contain no positive example



At 0.05% positives:

A batch of 512 has only 0.256 positives on average.

Most such updates contain only negative examples.

Larger or stratified batches improve exposure; they do not create new evidence.

Early training can mostly learn “make scores smaller”

9,995 negative events

Start with scores near 50%.

Together, negatives strongly push the shared output bias downward.

5 positive events

Each positive pushes upward.

There are very few examples from which to learn a generalizable positive pattern.

As scores fall, easy-negative gradients shrink. This is not an unavoidable zero-loss trap.

Learning the overall rarity can be much easier than learning who is more likely.

Oversampling makes positives visible — not more diverse

Original data

5 distinct positives

A few examples of what a detection looks like.

Oversampled batches

The same 5 positives appear repeatedly.

More training exposure.

Possible failure

The model memorizes those examples.

Unseen positives remain difficult.

More repetitions help optimization; more independent positives help generalization.

Changing the class balance changes what scores mean

Toy example: the features contain no information about the label.

Natural data

0.05%

Best constant BCE score

5% positive batches

5%

100× the population rate

25% positive batches

25%

500× the population rate

Better positive recall can come with an upward bias in predicted detection rates.

Reweightings also changes the training objective

What it does

A positive mistake receives a larger penalty.

This can encourage the network to learn rare positive patterns.

Why it can disappoint

A few positives can drive noisy updates or overfitting.

The score is no longer automatically a natural detection probability.

Weighting changes importance; it cannot add missing information.

Focal loss can focus on ambiguity as well as signal

Useful when...

Many easy negatives consume training attention.

Reducing their influence makes harder examples matter more.

Can fail when...

“Hard” means noisy or intrinsically unpredictable.

The model spends capacity on examples with little learnable signal.

Source: Lin et al. (2017). Optical-map implications are hypotheses to test.

A different loss cannot turn random outcomes into predictable events.

Mixup adds interpolation — not independent detections

Positive (label 1) + negative (label 0) → synthetic event (label λ)

Why it can help

Smoother predictions between examples; less memorization of individual training events.

Why it can fail

Interpolated features may not represent real events. Forced positive-negative pairing changes the label distribution.

If every pair is +/- and mean $\lambda = 0.5$, the synthetic mean label is 50%.

Natural pairing preserves the mean in expectation, but rarely includes positives.

Some events may be impossible to classify reliably

Learnable structure

Position or event features change the chance of detection.

The network can learn which events are more likely.

Unobserved randomness

Similar recorded inputs can still produce different outcomes.

No loss or sampler can reveal information absent from the inputs.

Accurate probabilities can be achievable even when individual hit prediction is weak.

For the optical map, protect the population mean

Training

Improve exposure to positives.

Retain a loss on naturally sampled real events.

Tune augmentation strength.

Validation

Use the natural class balance.

Hold out whole voxels.

Compare several seeds.

Measure both

Event ranking:
precision-recall / AP

Population map:
mean bias, MAE and
spatial trend

Success means useful ranking AND trustworthy rates; accuracy alone proves neither.

What would most directly improve the evidence?

At a true detection probability of 0.05%:

1,500 primaries

0.75 expected hits

47.2% chance of
zero hits

5,000 primaries

2.5 expected hits

8.2% chance of zero
hits

20,000 primaries

10 expected hits

<0.01% chance of
zero hits

More primaries reveal rare detections; more voxel locations reveal spatial structure.

Sampling tricks redistribute the evidence. Simulation supplies new evidence.

Appendix: why weights and sampling bias probabilities

Positive weight w changes the BCE optimum:

$$q^* = w p / (1 - p + w p)$$

At $p = 0.05\%$: $w = 100$ gives $q^* \approx 4.76\%$; $w = 1,999$ gives $q^* = 50\%$.

Under pure class-dependent resampling:

$$\text{logit}(p) = \text{logit}(q) + \text{logit}(\pi) - \text{logit}(r)$$

π : natural positive fraction; r : sampled fraction; q : calibrated sampled score.
Requires unchanged within-class distributions; not a general mixup correction.

Correct sampling weights can restore the original objective; arbitrary class weights do not.

References and scope

Lin et al. (2017) — Focal Loss for Dense Object Detection
openaccess.thecvf.com/content_iccv_2017/html/Lin_Focal_Loss_for_ICCV_2017_paper.html

Zhang et al. (2018) — mixup: Beyond Empirical Risk Minimization
<https://arxiv.org/abs/1710.09412>

Menon et al. (2021) — Long-tail learning via logit adjustment
<https://research.google/pubs/long-tail-learning-via-logit-adjustment/>

All numbers and diagrams here are illustrative calculations, not new training results.
Failure modes describe possibilities, not inevitable outcomes of these methods.

Detailed assumptions and derivations are included in the PowerPoint speaker notes.